

Chinh Dang

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EDUCATION

Worcester Polytechnic Institute (WPI), M.S in Computer Science | Worcester, MA

Jan 2026

- **GPA:** 3.9
- **Relevant Courseworks:** Design of Software System, Algorithm Design and Analysis, Database Management System, Software Security Design and Analysis, Big Data Management, Programming Language Design, Foundation of CS.

British University Vietnam (BUV), B.S in Computer Science | Hanoi, Vietnam

Oct 2023

- **GPA:** 3.7 | **Classification:** Second Class: First Division
- **Relevant Courseworks:** IT Infrastructure Security, Cyber Security, Operating System Internals and Biometrics, Ethical Hacking, Digital Technologies, Networking Concepts, Software Development, Web Development, Commercial Computing.

WORK EXPERIENCE

Web Developer (Part - Time) | 95 Hang Bong JSC - Hanoi, Vietnam

Jan 2023 – Dec 2023

- Developed and maintained a WordPress-based e-commerce website, improving user engagement through optimized product displays and up-to-date information.
- Managed and updated a MySQL database containing 1,000+ products by implementing batch update processes for inventory and metadata, ensuring data consistency while reducing manual effort and updating errors.
- Contributed to the transition and migration of a legacy PHP-based system to a Java Spring Boot architecture, assisting with code refactoring, feature parity validation, and integration testing to improve scalability, maintainability, and long-term system performance.

Software Engineer (Intern) | Meritel - Hanoi, Vietnam

June 2024 – Aug 2024

- Maintained, refactored, and enhanced backend modules of a production web application using Spring Boot and MongoDB, improving code readability, modularity, and long-term maintainability.
- Designed and implemented RESTful API endpoints to support core application features and integrations with third-party services, following REST best practices.

PROJECTS

- **Media Storage/Streaming Platform (v1):** Designed and implemented a modular, multi-service media streaming platform like Netflix using Spring Boot. Built secure authentication, media ingestion, and metadata management backed by PostgreSQL, Redis, and MinIO. Implemented on-demand, multi-resolution HLS streaming, where videos are live-transcoded from original media into in-memory HLS chunks and streamed immediately to users via a nginx/OpenResty edge, minimizing disk I/O and startup latency. Integrated event-driven content uploading and search indexing with Kafka and OpenSearch, and built background FFmpeg workers for transcoding, thumbnails, and cache cleanup. Designed the system with clear service boundaries, asynchronous workflows, and RAM-backed streaming for high-performance delivery. Support videos, images, and albums uploading with directory and filename management like cloud storage.
- **Emergency Vehicle Detection System:** Developed and deployed a YOLOv8-based computer vision system to detect emergency vehicles. Integrated custom image classification to identify active emergency states and used Raspberry Pi GPIO to control LED-based traffic lights. Enabled automatic traffic-light prioritization, improving simulation response time and traffic-flow modeling.
- **Automation Desktop Application:** Built a JavaFX desktop application to automate user-defined sequences of mouse and keyboard actions. Implemented user-friendly defined conditional checks and safety controls to ensure reliable execution. Improved user productivity by offloading repetitive tasks through configurable automation flows.
- **Product Catalog Website:** Created a dynamic product listing platform using Spring Boot, PostgreSQL, Tailwind CSS, and JavaScript. Implemented authentication and role-based authorization for secure product management. Supported category-based browsing and real-time updates to improve usability and maintainability.
- **Study Spot Mobile Application:** Developed an Android application using Kotlin and Jetpack Compose and integrating Google Maps SDK for real-time location tracking. Implemented geofencing to recommend nearby study locations automatically. Used Firebase Firestore for real-time synchronization, supporting multiple concurrent users.

SKILLS

- **Languages:** Java (Spring Boot, JavaFX, Java EE, Spring), JavaScript, Python, Kotlin, PHP
- **Frameworks & UI:** Spring Boot, Hibernate/JPA, Mockito, HTML, CSS, Tailwind CSS, Jetpack Compose, Thymeleaf
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, MinIO (S3-compatible)
- **Streaming & Search:** FFmpeg, HLS, OpenSearch, Kafka/Redpanda, Redis
- **Web & DevOps:** nginx/OpenResty, Docker, Docker Compose, Git, Linux, CI/CD, Jenkins
- **Cloud & Concepts:** AWS (EC2, Lambda, S3), Firebase, REST/SOAP APIs, JWT/OAuth2, event-driven systems, async processing, Linux, MacOS, Windows
- **Familiar With:** C, Rust, Hadoop, Spark, Apache Pig
- **Security:** JWT, OAuth2, authentication/authorization, validation, rate-limiting.

CERTIFICATES: Amazon Junior Software Developer, Introduction to Software Engineering